

Series and infinite products – 1

Convergence: the standard scale and comparison

Homework

12 сентября 2026 года

Theoretical minimum. Comparison and limit comparison work for eventually one-signed terms and are applied against the scale $\sum n^{-p}$, $\sum (n \ln^p n)^{-1}$, $\sum (n \ln n \ln^p \ln n)^{-1}$, each convergent exactly for $p > 1$. Condensation replaces $\sum a_n$ by $\sum 2^k a_{2^k}$ for monotone terms; the integral test replaces it by an integral and gives the error term as well. When the ratio tends to 1, expand it and use Raabe's test $\frac{a_{n+1}}{a_n} = 1 - \frac{p}{n} + O(n^{-2})$, or Gauss's form of it. Before comparing, check that the terms really are of one sign: for sign-changing series an asymptotic equivalence proves nothing. A useful reflex: a series whose terms are given by an integral, a recurrence or an arithmetic function is first replaced by its asymptotics, and only then compared.

1. Prove Raabe's test, and use it to decide the convergence of $\sum \left(\frac{(2n-1)!!}{(2n)!!} \right)^p$ for all real p . Show that the ratio test gives 1 here for every p .
2. Let $a_n \geq 0$ and let $\sum a_n$ converge. Prove that $\sum \sqrt{a_n a_{n+1}}$ and $\sum \frac{\sqrt{a_n}}{n}$ converge, give an example in which $\sum \sqrt{a_n}$ diverges, and show that for every $p \leq \frac{1}{2}$ the series $\sum \frac{\sqrt{a_n}}{n^p}$ may diverge.
3. (МФТИ 2006) Let $\alpha \in (0, 1)$. Suppose that the series $\sum \frac{x_k}{k^\alpha}$ converges and $x_k \geq x_{k+1} \geq 0$ for all k . Prove that the series $\sum x_k^{\frac{1}{1-\alpha}}$ also converges.
4. (IMC 2015) Prove that $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}(n+1)} < 2$.
5. (SEEMOUS 2022) For $p \in \mathbb{R}$, let $(a_n)_{n \geq 1}$ be the sequence defined by $a_n = \frac{1}{n^p} \int_0^n |\sin(\pi x)| x dx$. Determine all possible values of p for which the series $\sum a_n$ converges.

6. (SEEMOUS 2014) Consider the sequence (x_n) given by $x_1 = 2$ and $x_{n+1} = \frac{x_n+1+\sqrt{x_n^2+2x_n+5}}{2}$ for $n \geq 1$. Prove that the sequence $y_n = \sum_{k=1}^n \frac{1}{x_k^2-1}$ is convergent and find its limit.
7. (Высшая лига 2019) A sequence (x_n) with positive terms is monotonically increasing and bounded. Does the series $\sum \left(\sqrt{\frac{x_{n+1}}{x_n}} - \sqrt{\frac{x_n}{x_{n+1}}} \right)$ converge?
8. Let f be positive, continuous and non-increasing on $[1, \infty)$ with $\int_1^\infty f = \infty$. Prove that $\sum \frac{f(n)}{F(n)}$ diverges and $\sum \frac{f(n)}{F(n)^{1+\varepsilon}}$ converges for every $\varepsilon > 0$, where $F(x) = \int_1^x f$. (This is the continuous form of the Abel–Dini theorem; the discrete one is in the class sheet of Series and infinite products – 2.)
9. Decide the convergence of $\sum \frac{1}{n^{1+1/n}}$, of $\sum (n^{1/n} - 1)^p$ and of $\sum (\sqrt[n]{n} - 1)^n$, and explain which test is the cheapest in each case.
10. (SEEMOUS 2012) Let $a_n > 0$ for $n \geq 1$, and consider the right triangles $A_0A_1A_2, A_0A_2A_3, \dots, A_0A_{n-1}A_n, \dots$ arranged so that for every $n \geq 2$ the hypotenuse A_0A_n of the triangle $A_0A_{n-1}A_n$ is a leg of the next triangle $A_0A_nA_{n+1}$ with the right angle at A_n , the vertices A_{n-1} and A_{n+1} lie on opposite sides of the line A_0A_n , and $|A_{n-1}A_n| = a_n$ for every $n \geq 1$. Is it possible for the set $\{A_n\}$ to be unbounded while the series of angles $\sum_{n \geq 2} \angle A_{n-1}A_0A_n$ converges?